Palaeogeography, Palaeoclimatology, Palaeoecology, 24(1978): 263—277
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PALAEOMAGNETIC AGES OF LATE CRETACEOUS AND TERTIARY WEATHERED PROFILES IN THE EROMANGA BASIN, QUEENSLAND

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(Received March 7, 1977; revised version accepted July 7, 1977)

ABSTRACT

Idnurm, M. and Senior, B. R., 1978. Palaeomagnetic ages of Late Cretaceous and Tertiary weathered profiles in the Eromanga Basin, Queensland. Palaeogeogr., Palaeoclimatol., Palaeoecol., 24: 263—277.

Two iron-enriched weathered profiles, both widely distributed in the Eromanga and Surat Basins of Queensland, have been investigated palaeomagnetically. Normal and reversed polarities have been obtained from each profile, indicating that the weathering events spanned at least one reversal of the geomagnetic field. The minimum duration of weathering for the older of the two profiles is estimated as 10,000 years. The palaeomagnetic directions when compared with the Late Cretaceous and Cainozoic apparent polar wander curve for Australia give a Maastrichtian to Early Eocene age for one profile and an approximately Late Oligocene age for the other. These ages are consistent with times of high palaeotemperature indicated by oxygen isotope analysis of marine fossils of the Southeast Indian Ocean. The pole position for the younger profile is not significantly different from pole positions obtained in previous palaeomagnetic investigations of weathered rocks in Western Australia, South Australia and the Northern Territory, supporting the long-held view of a mid-Tertiary continent-wide weathering event in Australia.

INTRODUCTION

It has been recognised for a long time that a continent-wide zone of deep weathering exists in Australia affecting rocks of different types and ages. This zone was originally attributed to a single period of chemical weathering (Woolnough, 1927). In some regions more recent studies have shown it to comprise a number of distinct weathered profiles, suggesting a multicyclic origin, with phases of intense weathering alternating with periods of moderate or slight weathering (e.g. Doutch, 1976; Senior, in press). Although the Cainozoic weathering history of Australia appears to be more complex than initially thought, this does not necessarily preclude a common continent-wide series of weathering events. The problem has proved difficult to settle because of the lack of quantitative techniques for determining the ages of individual profiles, and for establishing regional and long distance correlations. Recent

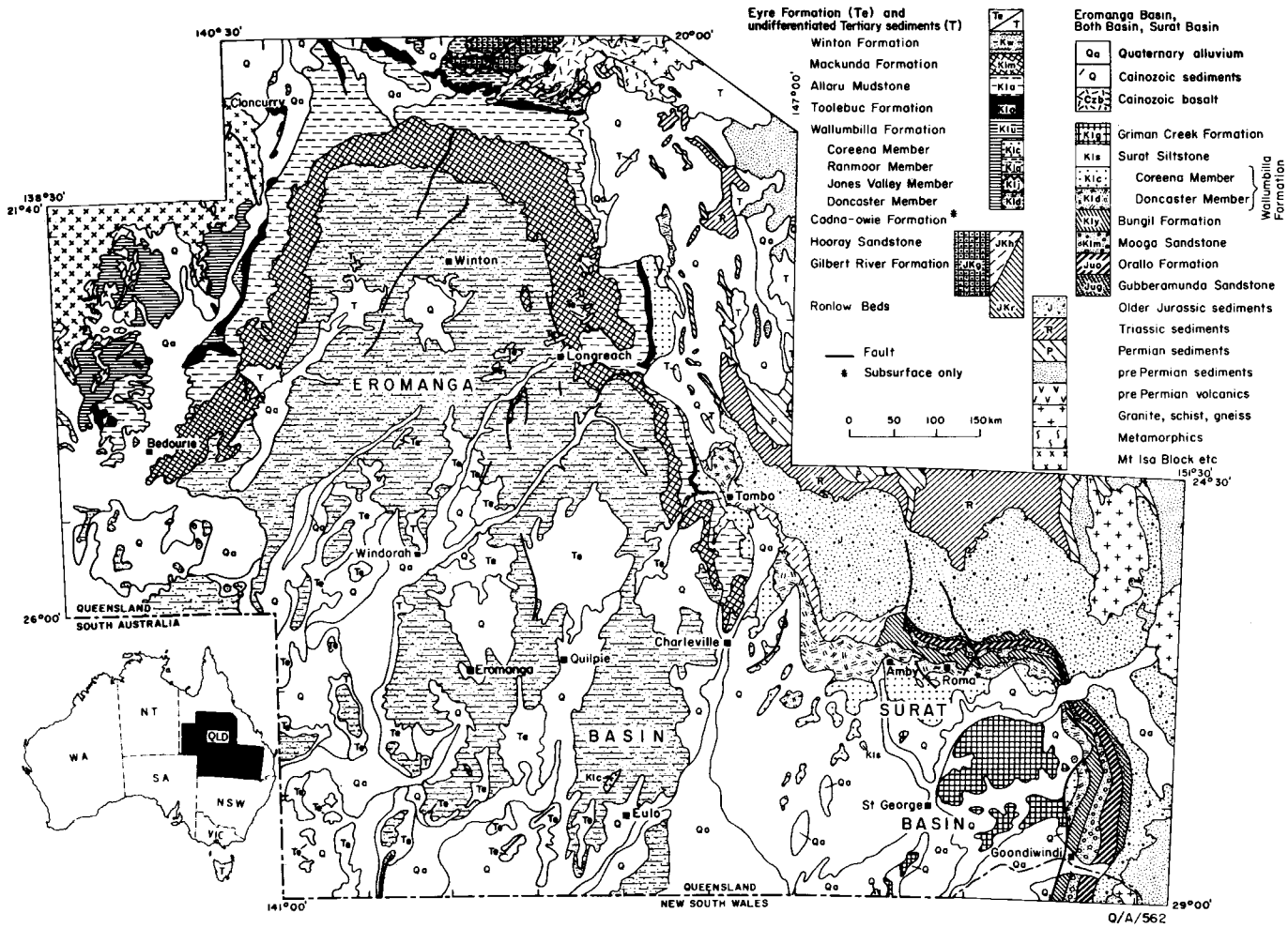


Fig.1. Geological map of the Eromanga and Surat Basins in Queensland (Modified from Exon and Senior, 1976).

applications of palaeomagnetism to the study of regional weathering in Western Australia (Schmidt and Embleton, 1976) and South Australia (Schmidt et al., 1977), suggesting a period of extensive mid-Tertiary weathering, have gone some way towards providing an answer.

The present work is concerned with the use of palaeomagnetic techniques for detailed regional weathering studies and extends the previous palaeomagnetic investigations of weathering to the Eromanga Basin of southwestern Queensland (Fig.1). Three weathered profiles have been identified in the Eromanga Basin, two in Cretaceous and one in Cainozoic host sedimentary rocks (Senior, 1976). Weathering of the Cretaceous rocks has given rise to thick kaolinitic profiles, while weathering of quartzose Cainozoic rocks has formed a thinner, though extensive, silcrete profile. At one time the kaolinitic profiles appear to have formed a weathered rock mantle which covered almost the entire southern part of the Eromanga and adjacent Surat Basins (Fig.2). The two kaolinitic profiles, referred to as the Morney and Canaway Profiles, are developed in quartz-poor sedimentary rocks of Early to Late Cretaceous age (Figs.1 and 2). The work reported here has investigated principally the palaeomagnetism of the ferruginous basal zone of the Morney Profile, and to a lesser extent the ferruginous upper zone of the Canaway Profile.

PARENT ROCKS AND MORPHOLOGY OF THE KAOLINITIC PROFILES

Detailed descriptions of the Cretaceous Winton and Mackunda Formation parent rocks of the weathered profiles of southwest Queensland have been given by Exon and Senior (1976). Briefly, these comprise a quartz-poor sequence of lithic and feldspathic sandstone, siltstone and mudstone. Montmorillonite is abundant and, together with lithic fragments, was probably derived from an andesitic volcanic terrain. The abundance of readily degradable minerals in the Cretaceous sequence appears to be at least partly responsible for the uniformity of the kaolinitic profiles developed in it.

The Eromanga Basin was tectonically stable during the Cainozoic, with only slight folding along predominantly meridionally trending axes (Senior et al., in press). The dips along the fold limbs are generally less than 3°, and rarely exceed 5°.

The Morney Profile (Senior, in Doutch, 1976) consists of up to 95 m of kaolinized sediments. All well-exposed sections of this profile can be subdivided into three zones of roughly equal thickness (Fig.3). At the base lies a ferruginous zone where the kaolinitic rocks are cemented by iron oxide and contain dense replacement ironstones in the form of lenses, beds and concretions. This zone is overlain by a vari-coloured kaolinized zone, which in turn is overlain by the uppermost zone composed of silicified rocks. The three zones have gradational boundaries between them. In most areas the Morney Profile underlies quartzose rocks of Cainozoic mixed fluviatile and lacustrine Eyre Formation with a shallow angular unconformity. The basal parts of the

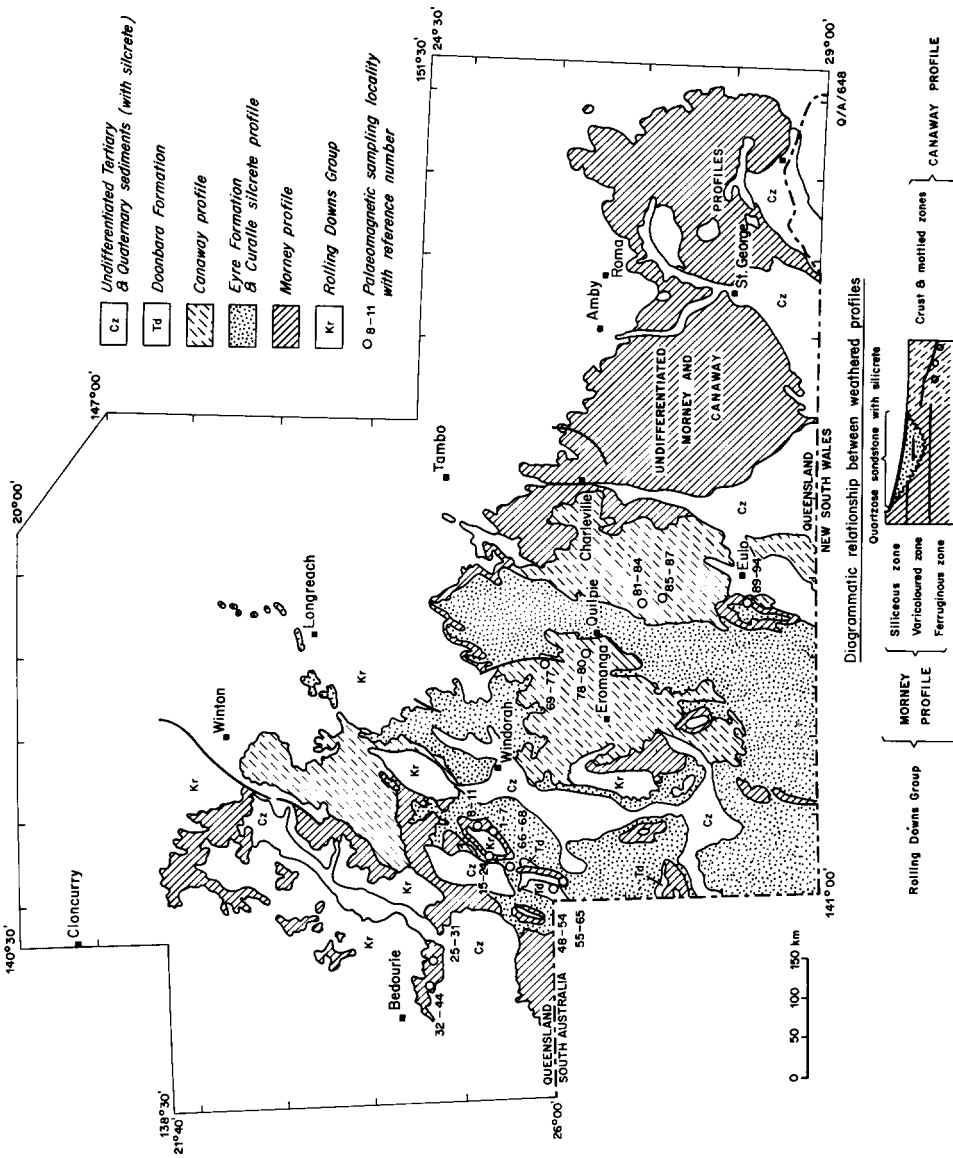


Fig. 2. Distribution of weathered profiles and palaeomagnetic sampling localities.

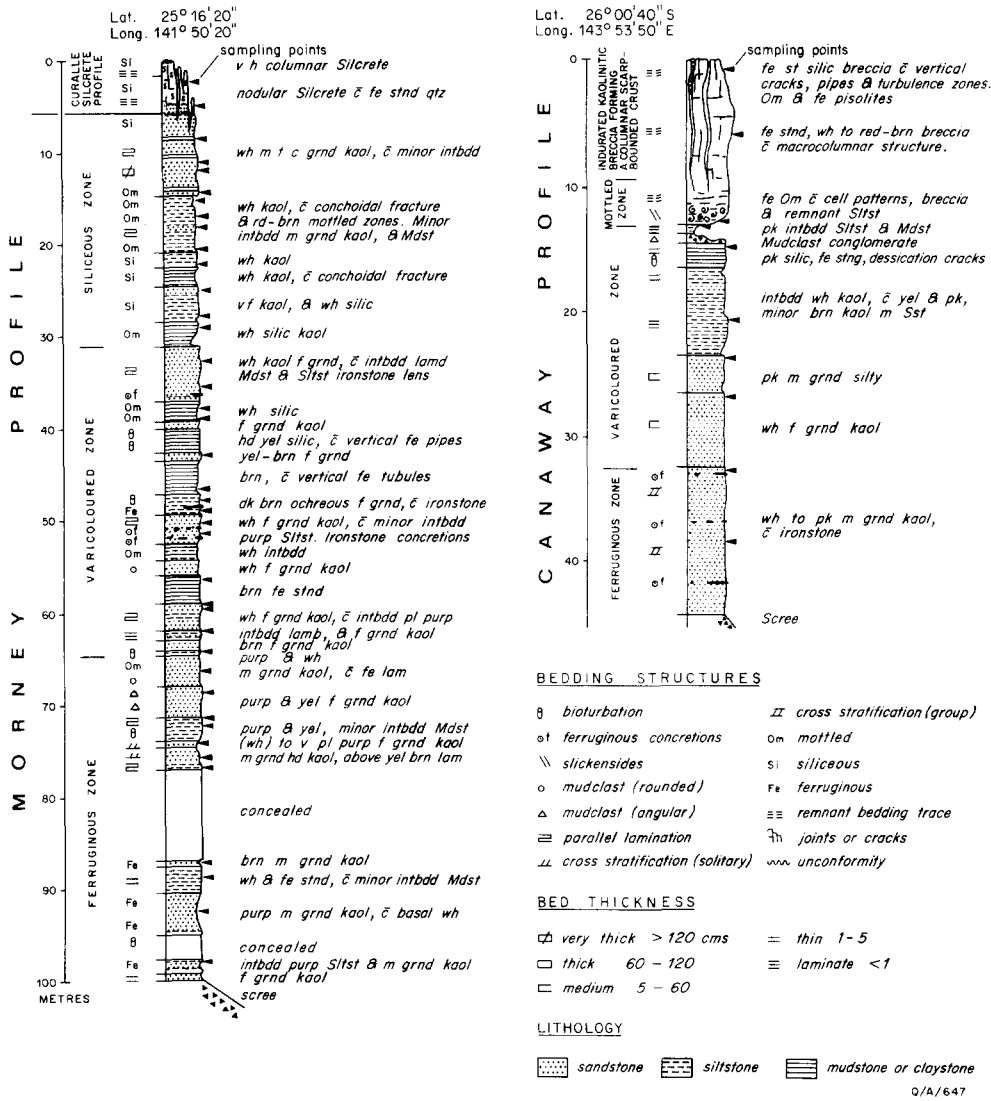


Fig.3. Reference sections of the Morney and Canaway Profiles.

Eyre Formation frequently contain clasts belonging to Morney Profile.

The Canaway Profile is capped by a distinctive crust of kaolinitic breccia impregnated with iron oxides which grades down through a zone of mottling into kaolinitic rocks of the unaltered Morney Profile (Fig.3). This profile, which has a present-day maximum thickness of 50 m, is considered to be younger than the Morney Profile, and has been postulated to have formed as follows (Senior, 1976): between the alluvial surfaces of the Eyre Formation there existed broad, low plains, consisting of soils and Morney Profile outcrop that had remained exposed to subaerial weathering. At these exposed locations

the uppermost parts of the Morney Profile were partly eroded, fragmented, and then further weathered during a second weathering cycle, forming the Canaway Profile. Although the parent rocks are extensively chemically altered their sedimentary structures were retained almost entirely (Fig.3).

GEOLOGICAL EVIDENCE OF THE AGE OF REGIONAL WEATHERING

The maximum age limit of the weathering is available from the age of the host rocks. Recently Burger and Kemp (in Senior, 1976) have identified Cenomanian palynomorphs in shallow stratigraphic drillholes in the Winton Formation, the youngest unit within the Eromanga Basin in which the kaolinitic profiles have developed. These microfossils occur within the upper part of the thickest, and hence probably the youngest, deposits of the Winton Formation in Queensland. A minimum age limit of approximately 23 m.y. has been established by isotopic dating of basalt flows which occupy erosional valleys in kaolinized weathered profiles (Exon et al., 1970). This investigation was carried out in the Roma-Amby area of Surat Basin (Fig.1), and although the weathering is continuous between the basins, it is not certain whether the basalt overlies the Morney or the Canaway Profile of the Eromanga Basin.

Another age estimate of the Morney Profile has been obtained by relating the processes of deposition of the Eyre Formation and regional weathering in southwest Queensland (Senior, 1976). The Eyre Formation is dated as Palaeocene/Eocene from microfossils at its depocentre, the Lake Eyre Basin (Wopfner et al., 1974). In some parts of the Eromanga Basin the Eyre Formation has been found to rest on unweathered Winton Formation. Wopfner et al. (1974) argue that this relationship results from the presence in the Early Tertiary of a standing body of water which prevented weathering. Support for this hypothesis comes from drillholes in southwestern Queensland (Senior, 1976) which show that kaolinitic weathered rocks thin progressively towards structural depressions and are absent below the Eyre Formation along synclinal axes. This implies that the weathering of the Cretaceous strata was partly contemporaneous with the sedimentation of the Eyre Formation within the broad and shallow structural valleys. Thus this indirect argument suggests a Palaeocene/Eocene maximum age limit for the Morney Profile. Geological evidence is lacking for the age of the Canaway Profile.

SAMPLING AND LABORATORY MEASUREMENTS

Seventy oriented hand samples of ironstone were collected from the basal ferruginous zone of the Morney Profile, and seventeen from the upper ferruginous zone of the Canaway Profile. The samples were collected along a traverse extending northwest from Eulo. The separation between the furthermost sampling localities of the Morney Profile was approximately 630 km; the corresponding separation for the Canaway Profile localities was 130 km. The sampling localities are shown in Fig.2 and the sample co-ordinates are

listed in Table I. Samples 34 to 45 have Mackunda Formation parent rock, all other samples are from the Winton Formation.

A PAR SM1 spinner magnetometer was used for the remanence measurements. A minimum of three specimens were prepared from each sample, and one or more of these was subjected to stepwise thermal demagnetization up to 660°C to determine the optimum temperature for removal of unstable components (Irving et al., 1961). The remaining specimens were thermally cleaned from unstable components at the determined optimum temperatures between 200°C and 300°C. Magnetic susceptibilities were measured after each demagnetization treatment in order to check that magnetic minerals remained chemically stable during heating. Except for a 5—20% decrease in susceptibility between 250°C and 300°C (and a corresponding decrease of up to 5% in remanence intensity) no changes were observed. This decrease probably results from the conversion of a small maghemite component in the ironstones to hematite, although the temperature is somewhat lower than usually reported in literature for this transition (see however Nagata, 1961). The decrease was not accompanied by any significant changes in the direction of magnetization.

RESULTS

(a) *Morney Profile*

The initial remanence (NRM) ranged from 10 to 1100 mA/m, depending mainly on the degree of secondary alteration by leaching of the ironstones. Secondary alteration was also found to have seriously affected the magnetic stabilities. The intensity of magnetization of a typical stable sample decreased slightly, including the small step decrease referred to above, as the sample was heated from room temperature to 650°C, and then decreased rapidly to small values between 650°C and 660°C. Fig.4 shows typical demagnetization curves for the ironstones.

Thirty samples were rejected because their magnetization directions remained random after cleaning, and three samples were rejected because their magnetization directions deviated by more than 30° from the mean direction. The cleaned sample directions of the remaining 37 samples are listed in Table I, and plotted together with the NRM directions in Fig.5a. The cleaned directions are grouped tightly into two closely antiparallel clusters; normal polarity is indicated by open circles and reversed polarity by filled circles. Applying the F ratio test of Watson and Irving (1957), the reversed directions, after inversion, are found not to differ significantly from the normal directions at the 5% level.

In order to eliminate the bias due to palaeosecular variations, palaeomagnetic data are usually grouped for analysis by sites. In this context each "site" denotes a rock unit that has been magnetized in a small interval of time, e.g. a sedimentary layer defined by closely spaced bedding planes, or a

TABLE I

Sample locality co-ordinates and mean sample directions after thermal cleaning

Sample no.	Latitude (deg. S)	Longitude (deg. E)	Declination (deg.)	Inclination (deg.)
Morney Profile				
1	25°22.2′	141°48.9′	010.3	−68.9
2	25°22.2′	141°48.9′	201.2	55.3
3	25°22.2′	141°48.9	183.6	71.9
4	25°22.2′	141°48.9′	176.0	77.2
6	25°22.2′	141°48.9′	195.9	65.5
8	25°16.3′	141°50.3′	202.1	59.5
11	25°16.3′	141°50.3′	030.2	−68.7
13	25°22.7′	141°31.4′	066.4	−74.2
14	25°22.7′	141°31.4′	012.2	−73.2
15	25°22.5′	141°31.3′	013.9	−72.1
16	25°22.5′	141°31.3′	018.3	−68.7
17	25°22.5′	141°31.3′	028.3	−73.3
18	25°22.5′	141°31.3′	015.9	−67.1
19	25°22.3′	141°31.1′	017.0	−71.7
20	25°22.3′	141°31.1′	034.0	−62.7
21	25°22.3′	141°31.1′	019.5	−68.4
22	25°21.3′	141°31.5′	054.2	−71.4
24	25°21.3′	141°31.5′	239.0	66.2
25	24°31.3′	140°22.3′	200.4	63.4
26	24°31.3′	140°22.3′	192.2	61.1
27	24°31.3′	140°22.3′	210.9	69.9
29	24°31.3′	140°22.3′	222.6	66.8
31	24°31.3′	140°22.3′	210.4	70.1
35	24°27.8′	140° 4.7′	196.8	77.0
58	26°12.2′	141°10.8′	352.9	−65.8
59	26°12.2′	141°10.8′	008.9	−69.5
60	26°11.7′	141°11.2′	029.1	−73.4
63	26°11.7′	141°11.4′	011.9	−68.2
64	26°11.7′	141°11.4′	007.6	−66.3
66	25°38.7′	141°20.7′	014.7	−72.9
67	25°38.7′	141°20.7′	019.1	−67.1
68	25°38.7′	141°20.7′	017.7	−74.3
70	26°19.8′	143°49.2′	202.5	69.9
71A	26°19.8′	143°49.2′	182.4	71.2
91	28°18.7′	144°37.7′	182.0	50.1
92	28°18.7′	144°37.7′	187.6	53.1
93	28°18.7′	144°37.7′	171.3	54.9
Canaway Profile				
72	26°19.8′	143°49.2′	011.4	−63.5
78	26°39.0′	144° 0.0′	201.0	60.5
79	26°39.0′	144° 0.0′	359.8	−61.2
80	26°39.0′	144° 0.0′	191.8	54.7
81	27°11.0′	144°24.0′	027.9	−52.1

TABLE I (continued)

Sample no.	Latitude (deg. S)	Longitude (deg. E)	Declination (deg.)	Inclination (deg.)
82	27°11.0′	144°24.0′	347.7	−46.3
84	27°11.0′	144°24.0′	004.5	−52.5
85	27°33.0′	140°30.0′	002.2	−65.6
86	27°33.0′	140°30.0′	016.2	−67.4
87	27°33.0′	140°30.0′	021.0	−62.0

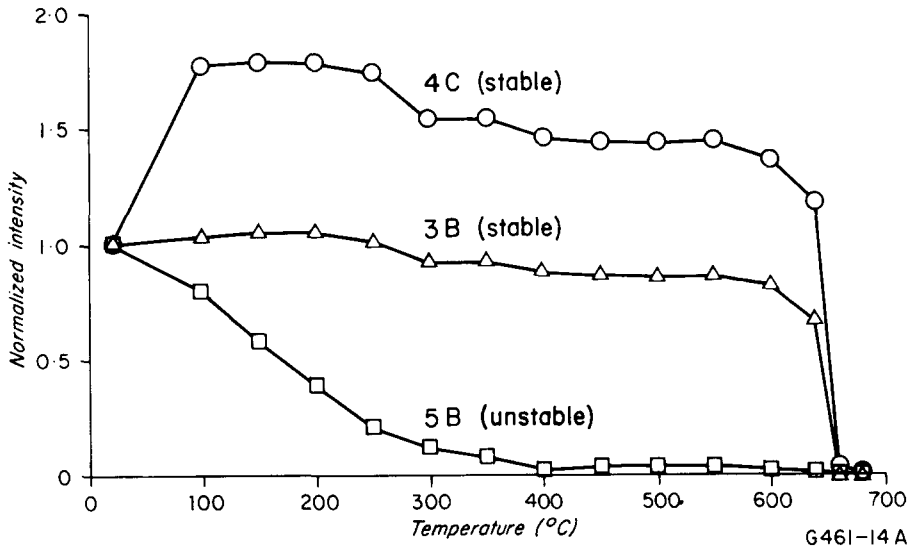
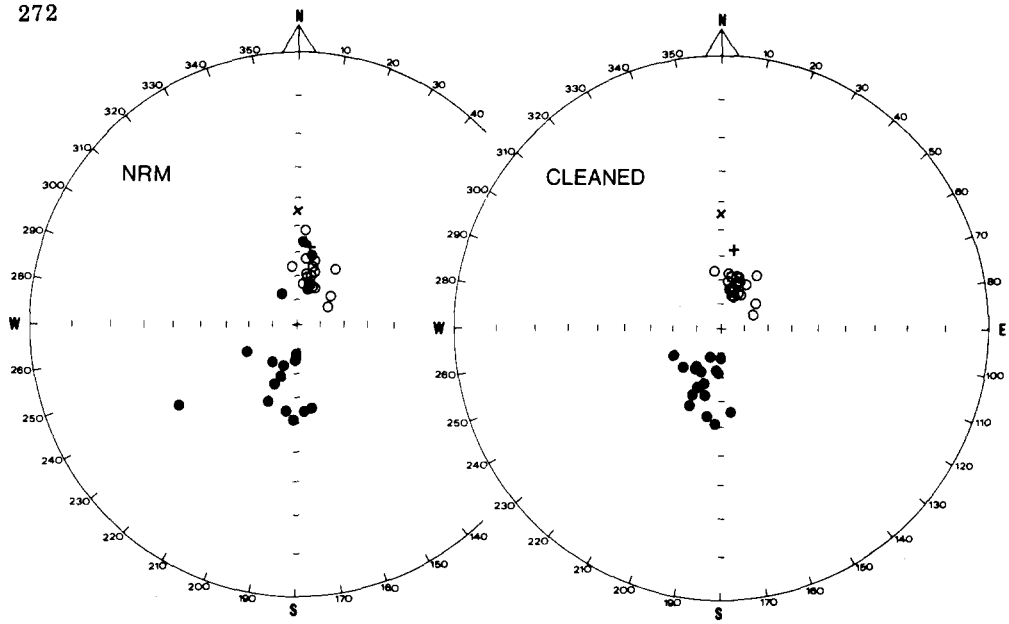


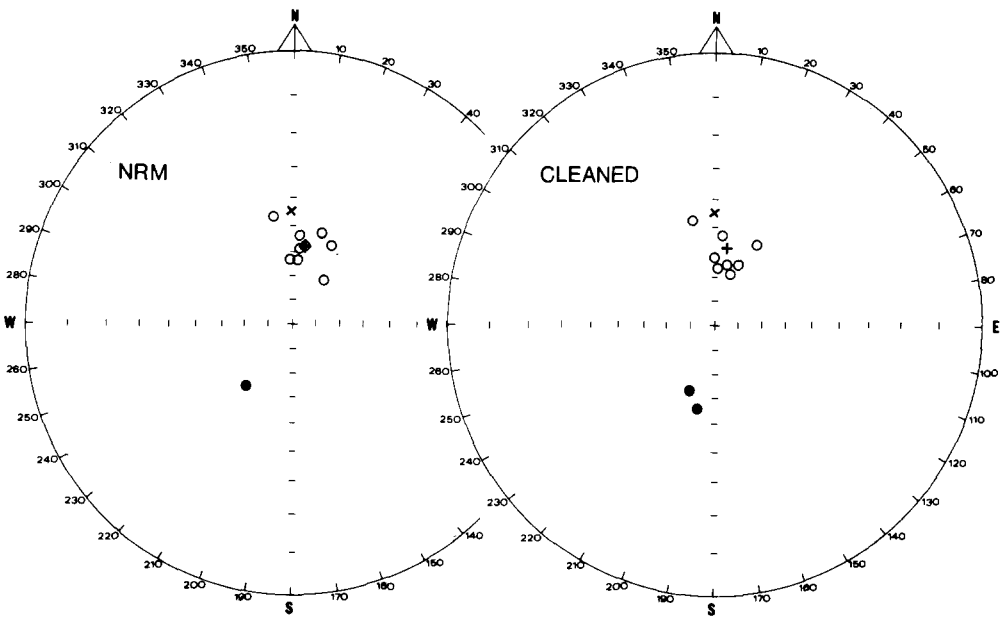
Fig.4. Typical thermal demagnetization curves for stable and unstable Morney Profile samples.

lava flow. If the sites span a sufficiently long time interval palaeosecular variations average out. For weathered profiles where distinct time markers such as bedding planes are not available, the elimination of secular variation bias presents special problems, for even widely separated locations do not necessarily represent different "sites". It is noted that the Morney Profile contains at least one polarity reversal, hence the time span for the development of this profile can be assumed to have exceeded the period of secular variations. Also, a polarity change was observed within three localities, and in particular a reversal was found within the same ironstone body in the locality at latitude 25°22.2′S, longitude 141°48.9′E. This suggests that the ironstone bodies formed slowly relative to the period of secular variations, and that samples collected from different parts of the same body represent different intervals of time. Consequently in calculating the mean Morney Profile direction each sample was assigned a unit weight.

The mean magnetization direction of the Morney Profile, found by combining the normal and reversed directions listed in Table I, is declination 17.8°, inclination −68.3°. The semiangle of the cone of 95% confidence, α_{95}, is 2.4°. The latitude and longitude of the pole are 59.8°S and 118.50°E



(a) MORNEY PROFILE



(b) CANAWAY PROFILE

Fig. 5. NRM and thermally cleaned directions of stable samples. (a) Morney Profile. (b) Canaway Profile. Open circles denote normal polarities; filled circles denote reversed polarities. X, present dipole; +, present field.

respectively, and the semiangle of the cone of 95% confidence, A_{95}, is 3.8°. If unit weights were assigned to locality mean directions (giving unit weights to separate groups of normal and reversed directions at each locality) rather than sample means, this calculation gives the pole as latitude 57.8°S, longitude 116.2°E, semiangle A_{95}, 5.2°.

Figure 6a shows the Cainozoic apparent polar wander curve for Australia, adapted from McElhinny et al. (1974). The Morney Profile pole is plotted on this curve as *MP* and corresponds to approximately 60 million years (m.y.). An error of 10 m.y. is estimated from the radius of the error circle and the rate of pole movement. The age of magnetization therefore lies in the Maastrichtian/Early Eocene (Van Eysinga, 1975).

(b) *Canaway Profile*

The initial remanence intensities varied from 10 to 250 mA/m. Seven samples were found to have unstable directions of magnetization. The mean directions before and after thermal cleaning of the remaining ten samples have been plotted in Fig.5b and the cleaned directions are listed in Table I.

The Canaway Profile contains both normal and reversed directions, and a reversal occurs within the locality latitude 26°39.0′S, longitude 144°0.0′E. Using the same argument as for the Morney Profile, the Canaway Profile mean direction is obtained by giving a unit weight to each sample. The mean direction is then found as declination 9.8°, inclination −59.1°, semiangle of cone of 95% confidence, 5.2°. The corresponding pole lies at latitude 74.1°S, longitude 115.1° E, and has A_{95} equal to 6.6°. The Canaway pole has been plotted as *CP* in Fig.6a.

A comparison of poles *MP* and *CP* indicates that the Canaway Profile is considerably younger than the Morney Profile. However, because of the limited number of samples and the disposition of the three poles adjacent to *CP* only an approximate age of 30 m.y. (Late Oligocene), with an uncertainty of 15 m.y., may be assigned to the Canaway Profile.

DISCUSSION

Assuming that deep weathering and kaolinization of the weathered rocks proceeded at the same time as the movement and precipitation of iron-oxides within the basal parts of the Morney Profile, the Maastrichtian/Early Eocene age of the ferruginous zone is also the age of the deep weathering. The duration of the weathering events is difficult to estimate, as very large numbers of samples would be required to establish statistically that a spread of directions exists, even for time spans as great as 2 m.y. The presence of normal and reversed polarities indicates however that the weathering covered a period of at least one reversal of the geomagnetic field. A minimum duration of weathering for the Morney Profile may be estimated by assigning the greater than 30° divergence of the three Morney Profile samples to polarity transition. Assum-

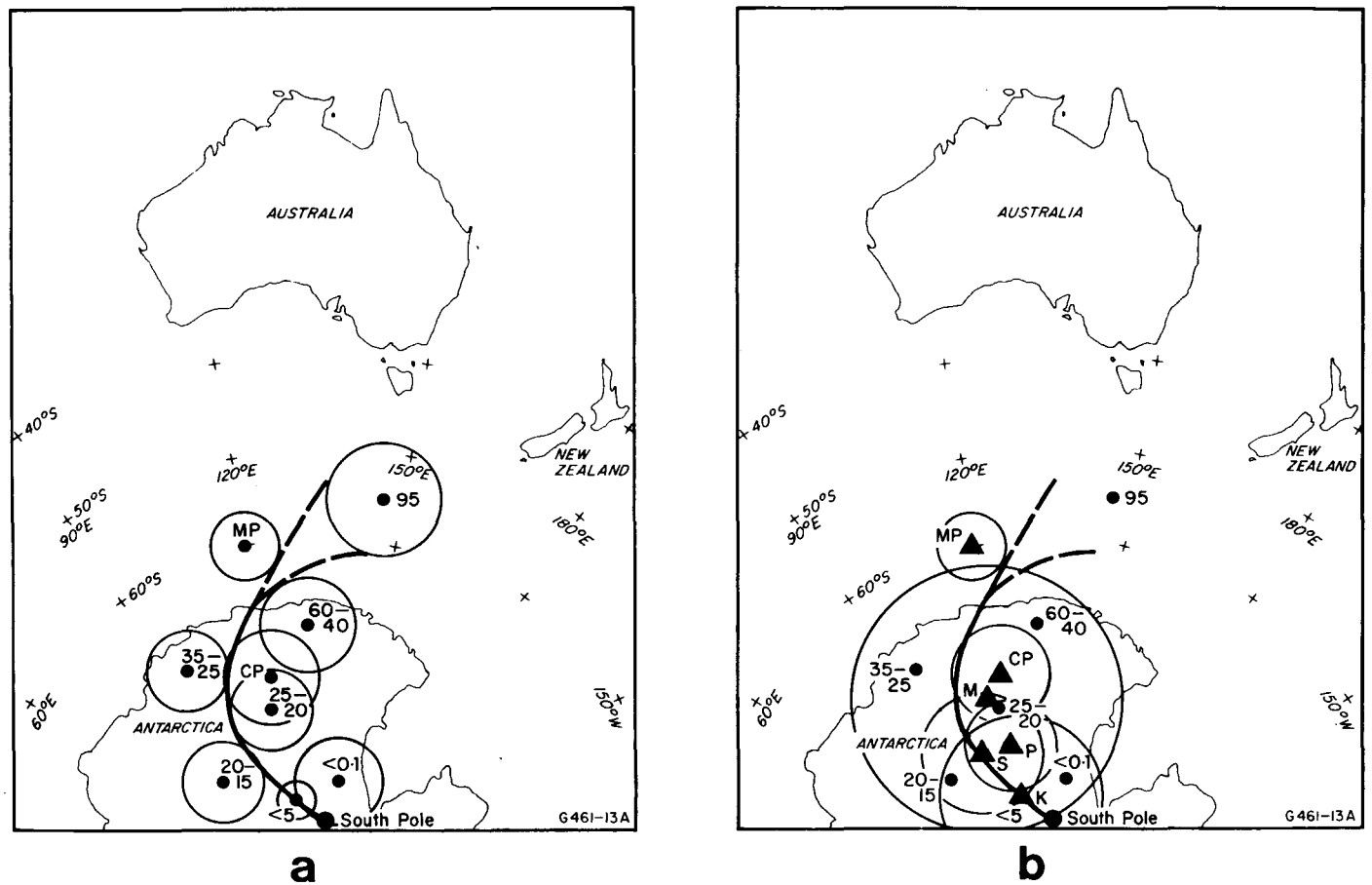


Fig.6.a. Morney Profile (*MP*) and Canaway Profile (*CP*) poles plotted on the Late Cretaceous and Cainozoic apparent polar wander path for Australia (adapted from McElhinny et al., 1974). b. Comparison of Australian Tertiary weathering poles. *P*, Perth Basin; *M*, Montejinni Limestone; *S*, Springfield Basin; *K*, Kangaroo Island.

ing a uniform rate of weathering, and a minimum of 1,000 years for polarity transitions (McElhinny, 1973), the lower limit for the duration of weathering is found to be 10,000 years. The palaeomagnetic age estimates confirm that, as postulated from field evidence, the Canaway Profile is significantly younger than the Morney Profile.

It is commonly believed that deep weathering requires warm and humid conditions, with possibly seasonal variations in precipitation (e.g. Goudie, 1973). The Cainozoic saw profound changes in climates in Australia resulting partly from changes in the ocean temperatures and partly from the separation and northwards drift of Australia from Antarctica (Kemp, 1978). Shackleton and Kennett (1975) have used O^{16}/O^{18} ratios of benthonic and planktonic foraminifera to establish a Cainozoic palaeotemperature curve of the South-east Indian Ocean. This curve shows very warm ocean temperatures, which are probably associated with warm and humid continental conditions, for Palaeocene to Early Oligocene. After the Early Oligocene surface temperatures decrease irregularly with decreasing age, with two brief reversions by about 3°C to warmer ocean conditions in the Early Miocene. The Early Miocene reversion agrees also with previous results of Dorman (1966), who determined palaeotemperatures from the oxygen isotope ratios of Tertiary fossils in Victoria. The palaeomagnetic age for the Morney Profile is consistent with the Palaeocene to Early Oligocene warm ocean temperatures around Australia; and the age of the Canaway Profile agrees within its error limits with the Early Miocene warming of the ocean.

It is interesting to compare the southwestern Queensland palaeomagnetic results with those reported recently from weathered sequences of the Perth Basin in Western Australia (Schmidt and Embleton, 1976), and from Kangaroo Island and the Springfield Basin in South Australia (Schmidt et al., 1977).

TABLE II

Comparison of pole positions from weathered rocks in Australia

Region	Pole latitude	Pole longitude	N	R	A_{95}	Reference
Perth Basin, W. Aust.	81.2°S	108.8°E	14	13.907	4.5°	Schmidt and Embleton (1976)
Montejinni Limestone, N.T.	76.4°S	107.2°E	12	11.039	14.1°	Luck (1970)
Springfield Basin, S. Aust.	80.4°S	90.5°E	6	5.96	5.6°	Schmidt et al. (1977)
Kangaroo Island, S. Aust.	86.9°S	76.9°E	8	7.92	8.4°	Schmidt et al. (1977)
Canaway Profile, S.W. Qld.	74.1°S	115.1°E	10	9.803	6.6°	This paper
Morney Profile, S.W. Qld.	59.8°S	118.5°E	37	36.001	3.8°	This paper

N = total number of sites/samples used in calculating the pole; R = resultant of N unit vectors; A_{95} = semiangle of the cone of 95% confidence.

Table II summarizes these results and includes also anomalous palaeomagnetic results from a Cambrian limestone in the Northern Territory, which are thought to originate from chemical remagnetization related to regional laterization (Luck, 1970). The pole positions have been plotted in Fig.6b.

It is seen from Fig.6b that the Canaway Profile pole lies close to the Montejinni Limestone pole and is not far removed from the poles for the Perth and Springfield Basins. The above four poles and the pole for Kangaroo Island form a group. The *F* ratio test of Watson and Irving (1957) shows that the five pole directions are indistinguishable at the 5% level. Thus palaeomagnetic evidence is consistent with widespread mid-Tertiary weathering in Australia, with a possible earlier initiation of this weathering in northern and northeastern regions compared to those of southern and western regions. The Morney Profile belongs however to a separate and altogether earlier period of weathering for which palaeomagnetism has not yet found any counterparts in other regions.

CONCLUSIONS

The age of the Morney Profile is estimated from palaeomagnetic data as Maastrichtian to Early Eocene, and the age of the Canaway Profile as Late Oligocene. The latter age has considerable uncertainty. Both ages are coeval with times of high temperatures of ocean waters around Australia obtained from oxygen isotope analysis of marine fossils. Each profile contains normal and reversed polarities, and has therefore formed over a period of time covering at least one reversal of the geomagnetic field. The minimum duration of weathering for the Morney Profile is estimated as 10,000 years.

The Canaway Profile pole lies very close to the pole position of chemically remagnetized Montejinni Limestone of the Northern Territory, and agrees well with pole positions for weathered sediments both in the Perth Basin, Western Australia, and in the Springfield Basin, South Australia. These four pole positions, together with a pole obtained for a laterite from Kangaroo Island, South Australia, form a statistically indistinguishable group.

Thus palaeomagnetic evidence indicates two periods of chemical weathering in southwest Queensland and supports the long-standing belief of a widespread mid-Tertiary weathering episode in Australia.

ACKNOWLEDGEMENTS

Dr P. Wellman carried out measurements on Morney Profile ironstone samples for one of us (BRS) in 1970, which demonstrated the feasibility of palaeomagnetic study of weathering in southwestern Queensland. We wish to thank him for these measurements and for valuable discussions, Drs E. Kemp and R. Jensen for discussion and criticism of the manuscript, P. Schmidt et al. for sending us a pre-publication copy of their paper, and R. Eaton for preparation of palaeomagnetic samples. The Research School of Earth Sciences,

Australian National University, kindly allowed us to use their palaeomagnetic facilities for the measurements reported in this paper.

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